
Small Language Multi-agent Debate to Identify Safety Attacks

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Abstract

Large language models (LLMs) are increasingly deployed in open environments where malicious prompts or poisoned data can compromise their safety alignment. Existing defenses often rely on large, centralized models, which are expensive and opaque. In this proposal, I introduce a lightweight and interpretable approach, **Small-Language Multi-Agent Debate (SLMAD)**, where multiple small language models engage in structured debate to judge the safety of questions and question-answer pairs. Each agent independently evaluates content, exchanges rationales, and reaches a consensus verdict on safety risk. This research aims to detect both unsafe prompts and harmful responses efficiently while maintaining explainability. This research will benchmark SLMAD against single-model classifiers and large-model judges on datasets derived from harmful and harmless instruction-response pairs, exploring its scalability, robustness, and interpretability in identifying safety attacks.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable generalization and reasoning capabilities across diverse domains such as education, healthcare, and software engineering [1, 2]. Despite their success, recent studies have highlighted significant safety risks associated with their open-ended generation abilities [3]. Malicious or ambiguous prompts can elicit harmful, biased, or privacy-violating responses, while adversarial fine-tuning or data poisoning can silently erode safety alignment. These vulnerabilities have raised serious concerns about the reliability of LLMs deployed in open and decentralized environments.

Recent work has revealed that safety alignment can be fragile both during training and inference. For example, recent studies such as [4] and [5] demonstrate that, in federated instruction tuning (FedIT), even a small fraction of malicious clients can inject unaligned or unsafe data that degrades global model safety without producing obvious deviations in gradient statistics or loss values. Post-hoc fine-tuning methods can partially restore alignment, but they do not *identify* unsafe inputs or poisoned client updates early in the process. Similarly, research on inference-time adversarial prompting and red teaming shows that large-scale alignment models can still be bypassed through subtle rephrasing, indirect jailbreaks, or social-engineering-style attacks [3].

Multi-agent debate has recently emerged as a promising paradigm for improving reasoning and factual reliability in LLMs [6]. By encouraging agents to critique and revise one another’s outputs, debate frameworks help expose inconsistencies and reduce model bias. However, existing debate methods typically rely on large foundation models and multi-round interactions, which are computationally expensive and difficult to deploy in constrained or federated environments.

To address these challenges, we propose a lightweight, interpretable, and scalable framework called **Small-Language Multi-Agent Debate (SLMAD)**. Instead of relying on a single large model or centralized safety classifier, SLMAD employs multiple small, heterogeneous language models

(SLMs) to collaboratively evaluate the safety of both questions and corresponding answers. Each agent independently determines whether a piece of content is safe, borderline, or unsafe, provides a concise rationale, critiques its peers’ reasoning, and then reaches a consensus verdict through structured debate.

The SLMAD framework is built upon three key insights. First, small language models can serve as efficient, transparent safety evaluators when organized collectively, avoiding the high cost and opacity of large-model judges. Second, debate-induced disagreement provides a natural signal of epistemic uncertainty, allowing ambiguous or adversarial cases to be escalated to human or large-model review. Third, by extending debate beyond questions to full question–answer (Q–A) pairs, SLMAD evaluates both the *intent* of the prompt and the *execution* of the response, identifying unsafe reasoning or enabling content even in superficially benign interactions.

This proposal aims to explore whether multi-agent debate among small models can provide a cost-effective and interpretable mechanism for detecting safety attacks and unsafe generations. We address the following research questions:

- Can multi-agent debate among small LMs reliably classify prompts and Q–A pairs as safe or unsafe compared to single-model safety classifiers or large-model judges?
- How does agent diversity, debate depth, and model size affect the accuracy, robustness, and interpretability of safety judgments?
- Can disagreement entropy and consensus scores serve as robust early-warning indicators of safety alignment drift or adversarial prompting attacks?

Through systematic experiments on synthetic and real-world safety datasets, including harmful/harmless instruction response pairs [4], BeaverTails, and AdvBench, this research will quantify SLMAD’s effectiveness, efficiency, and generalization. Ultimately, this research seeks to demonstrate that reliable safety evaluation does not require large centralized models; instead, coordinated debate among small agents can yield cost-efficient, transparent, and scalable defenses against emerging safety threats in large language models.

2 Related Work

2.1 Safety Alignment of Large Language Models

Ensuring that large language models behave safely and align with human values has become a central research goal in modern NLP. Early alignment work focused on human feedback and reinforcement learning from preference data, which trained models to follow user instructions while avoiding toxic or unethical behavior [1, 2]. More recent surveys provide a broad taxonomy of safety challenges, including jailbreak attacks, misinformation, hallucination, and bias [3]. These studies emphasize that safety is not a static property but rather a continual process of model auditing and adversarial evaluation.

2.2 Federated Instruction Tuning and Safety Attacks

Federated learning (FL) enables collaborative model training without direct data sharing, but it introduces new security and safety vulnerabilities. [7] summarize data-poisoning and model-manipulation attacks in FL, showing that even minor adversarial perturbations can corrupt the global model. Recent research has extended these risks to the domain of instruction-tuned LLMs. [4] demonstrate that malicious clients participating in federated instruction tuning (FedIT) can inject unaligned or unsafe data that silently deteriorates safety alignment, producing models that generate harmful content while remaining indistinguishable in gradient space. [5] further benchmarked multiple aggregation and defense methods, revealing that standard robust-FL techniques such as Krum or Trimmed Mean fail to mitigate these subtle safety attacks. Collectively, these studies show that federated ecosystems require active, semantic-level monitoring in addition to numerical anomaly detection.

2.3 Adversarial Prompting and Jailbreak Detection

Even when trained on aligned data, deployed models remain vulnerable to inference-time attacks. Adversarial prompts or indirect jailbreaks can cause aligned models to bypass safety filters or

to produce step-by-step harmful instructions [3]. A wide variety of automated red-teaming and evaluation frameworks have been proposed to expose such vulnerabilities, but most rely on large-scale models or heuristic classifiers. This motivates the need for lightweight, interpretable systems capable of judging both prompts and model outputs for safety without depending on massive foundation models.

2.4 Multi-Agent Debate and Collective Reasoning

Multi-agent debate frameworks have been introduced as a method for improving reasoning and truthfulness in LLMs [6]. In these systems, multiple agents independently answer a question, critique one another’s reasoning, and reach a consensus through structured argumentation. Debate has been shown to reduce factual errors and hallucinations by encouraging self-correction through disagreement. However, prior work typically focuses on factual or logical correctness rather than safety evaluation, and most implementations depend on very large models that make them computationally expensive.

2.5 Positioning of This Work

The proposed **Small-Language Multi-Agent Debate (SLMAD)** framework integrates these threads by using multiple small language models as cooperative critics that judge the safety of both questions and question–answer pairs. Unlike prior FL defenses that focus on numerical gradient anomalies [4, 5], SLMAD operates at the semantic level, identifying unsafe content through argument diversity and consensus. Compared with large-model debate systems [6], proposed approach is lightweight, interpretable, and suitable for deployment in constrained or federated environments. This synthesis aims to advance the study of scalable, transparent, and cost-efficient safety evaluation for modern language models.

3 Proposed Methodology

3.1 Overview

I propose to develop a framework called **Small-Language Multi-Agent Debate (SLMAD)** to evaluate and identify safety issues in both questions and question–answer (Q–A) pairs. Rather than relying on a single large model or an external safety classifier, SLMAD will coordinate multiple small language models (SLMs) that act as independent agents engaged in a structured debate. Each agent will independently assess whether a given input is safe, borderline, or unsafe, provide a concise rationale, and then critique the reasoning of its peers before reaching a collective verdict. Through this design, I aim to transform safety evaluation into a collaborative reasoning process, leveraging disagreement among small models to approximate large-model judgment quality with lower computational cost.

3.2 System Architecture

The proposed framework will consist of three main components:

1. **Input Parser:** This module will receive either a question q or a Q–A pair (q, a) and format it into standardized prompts for each agent.
2. **Agent Committee:** I will select a set of N heterogeneous small language models $\{M_1, M_2, \dots, M_N\}$, each with slightly different alignment or reasoning profiles (for example, a safety-tuned model, a reasoning-oriented model, and a literal instruction-follower).
3. **Moderator:** I will design a lightweight meta-model to aggregate agent judgments, resolve disagreements, and produce a final *SafetyScore*.

By incorporating heterogeneity across agents, I expect the system to capture diverse reasoning patterns and reduce correlated biases. The moderator will serve as a simple, interpretable aggregation mechanism rather than a heavy decision-making model.

3.3 Debate Protocol

The debate process will follow two concise rounds:

1. **Round 1 (Independent Judgment):** Each agent M_i will receive the input $x \in \{q, (q, a)\}$ and generate a triplet (l_i, r_i, c_i) , where l_i is its safety label (*safe*, *borderline*, or *unsafe*), r_i is a short textual rationale, and c_i is a self-reported confidence score.
2. **Round 2 (Critique and Revision):** Each agent will review anonymized rationales from its peers and may revise its initial label and confidence, producing (l'_i, c'_i) . This step is intended to simulate peer review and promote self-correction among agents.

The moderator will then aggregate results using a confidence-weighted majority vote to generate an overall *SafetyScore*. Entropy of the label distribution will be used to measure disagreement among agents; higher entropy values will indicate uncertainty and trigger further review by a larger model or human evaluator.

3.4 Safety Taxonomy

The debate will be guided by a consistent taxonomy for labeling:

- **Question Safety (SQ):** *Safe*, *Borderline*, or *Unsafe*, where *unsafe* denotes explicit intent to cause harm or seek illegal information.
- **Q–A Pair Safety (QA):**
 1. *Harmless-Refusal*: An answer that safely rejects the unsafe query.
 2. *Safe-Information*: A factual yet non-enabling response.
 3. *Unsafe-Enabling*: A harmful or enabling response that violates alignment.
 4. *Unclear*: Ambiguous or low-confidence outputs requiring escalation.

This taxonomy will allow SLMAD to assess both the intent of user prompts and the safety of model responses, addressing gaps in traditional one-dimensional safety detection systems.

4 Planned Experiments and Evaluation

In this section, I outline the experimental design that I plan to conduct after developing the proposed **Small-Language Multi-Agent Debate (SLMAD)** framework. These experiments are intended to assess the effectiveness, interpretability, and efficiency of SLMAD in detecting safety issues across both questions and question–answer (Q–A) pairs. No experiments have been performed yet; the following subsections describe the setup that I intend to implement in the future.

4.1 Objectives

The main objectives of the planned experiments will be:

- To evaluate whether small-language multi-agent debate can classify questions and Q–A pairs as safe or unsafe with accuracy comparable to larger models.
- To analyze the relationship between agent diversity, debate rounds, and model size on overall detection performance.
- To examine how disagreement entropy and consensus confidence correlate with the uncertainty and difficulty of safety judgments.
- To quantify the trade-offs between computational cost and safety classification accuracy.

4.2 Datasets

I plan to use a mix of publicly available and synthetically generated datasets to train and evaluate SLMAD:

- **Harmful and Harmless Instruction–Response Pairs:** These will be generated following the approach described by [4], which categorizes data into aligned (*safe*) and unaligned (*unsafe*) subsets.
- **BeaverTails:** A benchmark containing diverse harmful prompts and benign counterparts, suitable for testing question-level safety detection.

- **AdvBench:** A large-scale dataset of adversarial prompts used to evaluate the robustness of LLM safety defenses.
- **Additional Synthetic Data:** I will generate new adversarial and borderline prompts using smaller models to simulate challenging safety cases and to improve coverage of ambiguous examples.

4.3 Baseline Methods

To evaluate the benefits of SLMAD, I plan to compare its performance with several baseline approaches:

- A single small safety classifier trained on the same data without debate.
- A medium-scale language model (e.g., 7B parameters) used as a direct safety judge.
- Rule-based or keyword-filter baselines that provide high precision but limited recall.
- Existing large-model debate frameworks [6] adapted for safety evaluation (if computationally feasible).

These baselines will help isolate the contribution of multi-agent debate and consensus mechanisms to overall safety performance.

4.4 Evaluation Metrics

The evaluation will focus on both classification performance and interpretability:

- **Accuracy and F1-score:** For safe vs. unsafe classification on both questions and Q–A pairs.
- **Unsafe Recall:** The proportion of genuinely unsafe cases correctly detected.
- **Disagreement Entropy:** A measure of inter-agent uncertainty, used to analyze the relationship between debate diversity and reliability.
- **Efficiency Metrics:** Inference latency and total token usage per item, to quantify the cost of running the debate compared to single-model evaluation.
- **Interpretability:** Qualitative analysis of rationale quality and clarity of consensus explanations.

4.5 Planned Ablation Studies

To better understand the impact of various design factors, I plan to perform several ablation studies:

- **Committee Size:** Varying the number of agents ($N = 2$ to 5) to observe the effect on accuracy and uncertainty.
- **Model Heterogeneity:** Comparing homogeneous (same model family) vs. heterogeneous (different architectures) agent committees.
- **Debate Depth:** Comparing single-round vs. two-round debates to measure the benefit of peer critique.
- **Confidence Calibration:** Testing different weighting and normalization schemes for agent confidence scores.

4.6 Expected Outcomes

I expect that SLMAD will:

- Achieve comparable or superior unsafe detection performance relative to single small-model baselines.
- Demonstrate that disagreement entropy can serve as a reliable signal for uncertain or adversarial inputs.
- Offer interpretable safety assessments through rationale summaries and consensus explanations.

- Operate at a significantly lower computational cost than large-model debate systems.

These outcomes would support the central hypothesis that coordinated reasoning among small models can provide an efficient and explainable mechanism for detecting safety attacks in large language models.

4.7 Future Work Beyond Initial Evaluation

Depending on the success of initial experiments, I plan to extend this work by integrating SLMAD into federated instruction-tuning pipelines to detect and isolate unsafe client contributions.

5 Next Steps

References

- [1] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, et al. Training language models to follow instructions with human feedback. *arXiv preprint arXiv:2203.02155*, 2022.
- [2] Yuntao Bai et al. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*, 2022.
- [3] Richard Pang, Yixuan Wang, et al. A survey on safety alignment of large language models. *arXiv preprint arXiv:2403.05618*, 2024.
- [4] Rohit Gupta, Rui Ye, Jingyi Chai, Xiangrui Liu, Yanfeng Wang, and Siheng Chen. Emerging safety attack and defense in federated instruction tuning of large language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.
- [5] Rui Ye, Xiangrui Liu, et al. Openfedllm: Benchmarking federated instruction tuning for large language models. In *NeurIPS*, 2024.
- [6] John Chern, Yash Bhargat, Meera Madan, et al. Combating adversarial attacks with multi-agent debate. In *International Conference on Learning Representations (ICLR)*, 2024.
- [7] Lingjuan Lyu et al. A survey on data poisoning attacks and defenses in federated learning. *ACM Computing Surveys*, 2022.